Lab Manual: Practical 8

Course Code: ETCS406

Course Title: Game Development

Program: B. Tech Computer Science and Engineering

Semester: VII

Practical Title: Conduct User Testing for Your Game Prototype. Gather Feedback, Iterate, and Update Game Elements Accordingly. Document the Process.

Duration: 3 Hours

Lab Type: Evaluation & Iteration

Assessment Weightage: Part of CAL 1 (Lab Assignments)

Objective

To conduct structured user testing for a game prototype, gather qualitative and quantitative feedback, analyze results, and iterate on the game design to improve user experience. The process will be documented for evaluation and future reference.

Pre-requisites

- A working game prototype (2D/3D/AR/VR) developed in Unity
- Basic understanding of UX principles and playtesting methods
- Familiarity with Unity Editor and version control
- Exposure to feedback collection tools (forms, interviews, observation)

Learning Outcomes

By the end of this lab, students will be able to:

- Design and execute a user testing session for a game prototype
- Collect and analyze feedback from real users
- Identify usability issues and areas for improvement
- Modify game elements based on feedback and document the iteration process

Tools & Resources

- Unity Editor (latest version)
- Feedback forms or Google Forms
- Screen recording tools (e.g., OBS Studio)
- Spreadsheet or document editor for analysis
- Optional: Survey tools like Typeform or Microsoft Forms

Lab Tasks

Task 1: Prepare for User Testing

- Define testing goals (e.g., usability, engagement, difficulty)
- Select 3–5 users from target audience (peers or volunteers)
- Create a feedback form with questions on gameplay, controls, visuals, and overall experience
- Set up screen recording or observation tools

Task 2: Conduct Testing Sessions

- Brief users on the game and its objectives
- Observe users as they play the game prototype
- Record gameplay sessions and note user reactions
- Collect completed feedback forms or conduct short interviews

Task 3: Analyze Feedback

- Categorize feedback into themes: bugs, confusion, enjoyment, suggestions
- Identify common issues or patterns
- Prioritize changes based on impact and feasibility

Task 4: Iterate and Update

- Modify game elements (UI, controls, level design, mechanics) based on feedback
- Test updated version internally
- Document changes made and rationale for each update

Task 5: Final Documentation

- Create a report including:
    - Summary of testing goals and methods
    - User feedback highlights

- List of changes made
- Screenshots or video clips of before/after
- Reflection on the process and learning outcomes

Evaluation Criteria

Component	**Marks**
Testing Setup & Execution	3
Feedback Collection & Analysis	3
Game Iteration & Improvements	2
Documentation & Presentation	2
Total	**10**

Tips for Success

- Encourage honest and constructive feedback
- Avoid leading questions in your survey
- Focus on improving player experience, not just fixing bugs
- Keep documentation clear and visual for better impact

Suggested Reading & Resources

- *The Art of Game Design* by Jesse Schell – Chapters on playtesting and iteration
- Unity Learn: Playtesting and Feedback
- Gamasutra: Articles on user testing and game UX
- YouTube: Game UX tutorials by Game Maker's Toolkit

Submission Guidelines:

- Submit your report in PDF format via the Google Classroom.
- Include screenshots or gameplay references where relevant.